

Lecture 20: Change of Variables & the Jacobian

Background & Motivation

We've used polar, cylindrical, and spherical coordinates with extra factors of r or $\rho^2 \sin \phi$ in the integrand. Today we explain where those factors come from: the Jacobian determinant, which measures how a coordinate transformation stretches or shrinks area and volume. This also lets us design custom substitutions for specific regions.

1. Warm-Up: Where Does $r \, dr \, d\theta$ Come From?

In polar coordinates, $dA = r \, dr \, d\theta$. We have used this factor of r many times – today we explain *why* it appears.

Evaluate $\int_0^{2\pi} \int_0^3 r \cdot r \, dr \, d\theta$ to find the area of a disk of radius 3.

1. Inner (r): $\int_0^3 r \, dr = 9/2$ (using $f = 1$, so integrand is just r from dA)
2. Outer (θ): $2\pi \cdot 9/2 = 9\pi$
3. Check: $\pi(3)^2 = 9\pi \checkmark$

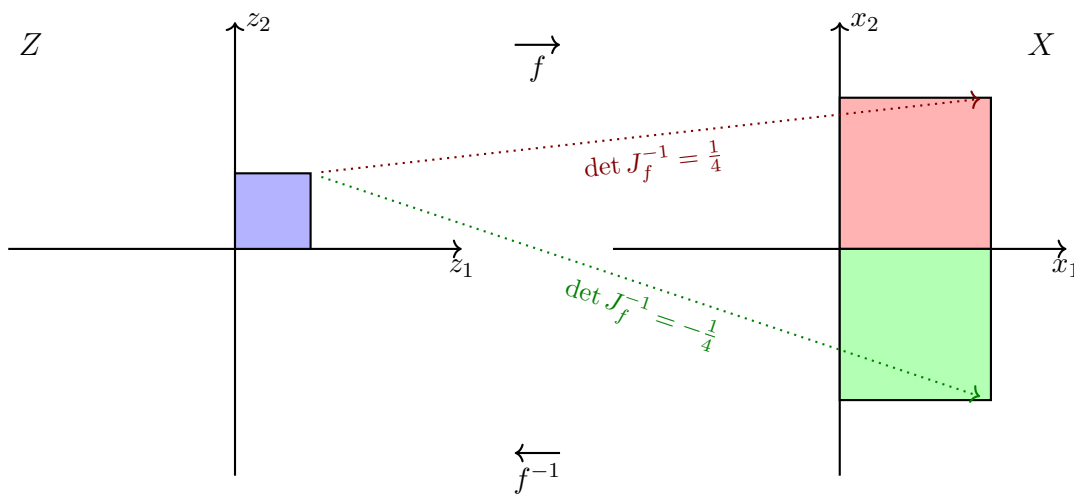
2. The Jacobian Determinant

Definition 1: Jacobian (2D)

For a transformation $T: (u, v) \mapsto (x, y)$, the **Jacobian** is:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

- $|J|$ measures how much the transformation stretches or shrinks area at each point
- $dA_{xy} = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv$



How the Jacobian determinant measures area scaling under a transformation. Different parts of the source region Z get scaled by different factors when mapped into X — the absolute value of $|J|$ gives the local scaling factor, and a negative determinant indicates a reflection.

Definition 2: Jacobian (3D)

For $(u, v, w) \mapsto (x, y, z)$:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

– Same idea: $dV_{xyz} = |J| du dv dw$

3. Change of Variables Formula

Theorem 1: Change of Variables (Double Integral)

$$\iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

where S is the region in the uv -plane corresponding to R .

Triple integral version:

$$\iiint_R f dV = \iiint_S f(x(u, v, w), \dots) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw$$

4. Examples

Example 1: Polar Jacobian Verification

Verify that the Jacobian of $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$ is r .

1. **Compute partials:** $\frac{\partial x}{\partial r} = \cos \theta$, $\frac{\partial x}{\partial \theta} = -r \sin \theta$

$\frac{\partial y}{\partial r} = \sin \theta$, $\frac{\partial y}{\partial \theta} = r \cos \theta$

2. **Determinant:** $J = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$

3. **Result:** $|J| = r$, confirming $dA = r dr d\theta$. $\boxed{J = r}$

Example 2: Custom Substitution for a Parallelogram

Evaluate $\iint_R (x - y) dA$ where R is the parallelogram with vertices $(0, 0)$, $(2, 1)$, $(3, 3)$, $(1, 2)$.

1. **Identify edges:**

- From $(0, 0)$ to $(2, 1)$: slope $1/2$, so $y = x/2$ i.e. $2y - x = 0$
- From $(1, 2)$ to $(3, 3)$: slope $1/2$, so $2y - x = 3$
- From $(0, 0)$ to $(1, 2)$: slope 2 , so $y = 2x$ i.e. $y - 2x = 0$
- From $(2, 1)$ to $(3, 3)$: slope 2 , so $y - 2x = -3$

2. **Choose transformation:** Let $u = 2y - x$ and $v = y - 2x$.

Bounds become: $0 \leq u \leq 3$, $-3 \leq v \leq 0$.

3. **Invert:** From $u = 2y - x$ and $v = y - 2x$:

$$u - 2v = 2y - x - 2y + 4x = 3x \implies x = \frac{u - 2v}{3}$$

$$2u - v = 4y - 2x - y + 2x = 3y \implies y = \frac{2u - v}{3}$$

4. **Jacobian:** $J = \begin{vmatrix} 1/3 & -2/3 \\ 2/3 & -1/3 \end{vmatrix} = \frac{-1}{9} + \frac{4}{9} = \frac{1}{3}$, $|J| = \frac{1}{3}$

5. **Integrand:** $x - y = \frac{u - 2v}{3} - \frac{2u - v}{3} = \frac{-u - v}{3}$

6. **Evaluate:** $\int_0^3 \int_{-3}^0 \frac{-u - v}{3} \cdot \frac{1}{3} dv du = \frac{1}{9} \int_0^3 \int_{-3}^0 (-u - v) dv du$

$$= \frac{1}{9} \int_0^3 \left[-uv - \frac{v^2}{2} \right]_{-3}^0 du = \frac{1}{9} \int_0^3 \left(-3u + \frac{9}{2} \right) du$$

$$= \frac{1}{9} \left[-\frac{3u^2}{2} + \frac{9u}{2} \right]_0^3 = \frac{1}{9} \left(-\frac{27}{2} + \frac{27}{2} \right) = 0$$

7. **Result:** $\boxed{0}$

(By symmetry of the parallelogram and the integrand, this makes sense.)

Example 3: Spherical Jacobian Verification

Show that the Jacobian for spherical coordinates is $\rho^2 \sin \phi$.

1. **Transformation:** $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$.

2. 3×3 **Jacobian matrix:**

$$J = \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta & \rho \cos \phi \cos \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta & \rho \cos \phi \sin \theta \\ \cos \phi & 0 & -\rho \sin \phi \end{vmatrix}$$

3. **Expand along Row 3:**

$$\begin{aligned} &= \cos \phi (-\rho^2 \sin \phi \cos \phi \sin^2 \theta - \rho^2 \sin \phi \cos \phi \cos^2 \theta) \\ &\quad + (-\rho \sin \phi)(\rho \sin^2 \phi \cos^2 \theta + \rho \sin^2 \phi \sin^2 \theta) \\ &= -\rho^2 \sin \phi \cos^2 \phi - \rho^2 \sin^3 \phi = -\rho^2 \sin \phi (\cos^2 \phi + \sin^2 \phi) \end{aligned}$$

4. **Result:** $J = -\rho^2 \sin \phi$, so $|J| = \boxed{\rho^2 \sin \phi}$

Example 4: Ellipse Area via Substitution

Use a change of variables to show that the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab .

1. **Substitution:** Let $x = au$, $y = bv$.

2. **New region:** The ellipse becomes $u^2 + v^2 = 1$ (the unit disk).

3. **Jacobian:** $J = \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab$, $|J| = ab$

4. **Area:** $A = \iint_R dA = \iint_{\text{unit disk}} ab \, du \, dv = ab \cdot (\text{area of unit disk}) = ab \cdot \pi$

5. **Result:** $A = \pi ab$

Practice Problems

Problem 1. Compute the Jacobian $|J|$ of the transformation $x = u + v$, $y = u - v$.

Answer: $|J| = 2$

Problem 2. Verify the cylindrical coordinate Jacobian: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$. Show $|J| = r$.

Answer: $|J| = r$

Problem 3. Use $u = x + y$, $v = x - y$ to evaluate $\iint_R (x^2 - y^2) \, dA$ where R is the square with vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$, $(0, -1)$.

Hint: $x^2 - y^2 = (x + y)(x - y) = uv$ and the region in uv -space is the square $|u| \leq 1$, $|v| \leq 1$.

Answer: 0

Problem 4. Use $x = 2u$, $y = 3v$ to evaluate $\iint_D xy \, dA$ where D is the region inside the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$, first quadrant only.

Answer: $\int_0^{\pi/2} \int_0^1 (2r \cos \theta)(3r \sin \theta) \cdot 6r \, dr \, d\theta = 36 \int_0^{\pi/2} \sin \theta \cos \theta \, d\theta \int_0^1 r^3 \, dr = 36 \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{9}{2}$

Problem 5. Let $T: (u, v) \mapsto (u^2 - v^2, 2uv)$. Compute the Jacobian and describe geometrically what this transformation does. *Hint:* think complex numbers, $z = u + iv \mapsto z^2$.

Answer: $J = \begin{vmatrix} 2u & -2v \\ 2v & 2u \end{vmatrix} = 4u^2 + 4v^2 = 4(u^2 + v^2)$